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Course 1

Course 1.1 ⇒ Introduction to Data Analysis

1.1.1 Computational Intelligence (CI)

Definition: **CI represents the capability of a computer to learn specific tasks from data or experimental observation**.
Core Principles:

- **Fuzzy Logic**: **Handles imprecise information and uncertainty**
- **Neural Networks**: Learns patterns from data through connectionist models
- **Evolutionary Computation**: Optimizes solutions through biologically inspired algorithms
- **Learning Theory**: Establishes foundations for data-driven model generalization
- **Probabilistic Methods**: Models uncertainty through probability distributions

CI vs. AI:

Both aim for general intelligence, however, **CI is a subset of AI**, **focusing on soft**, **human-like reasoning** rather than hard logic or symbolic approaches.

1.1.2 Data Analysis

Definition: **A systematic process of inspecting, cleaning, transforming, and modeling data to derive insights, support decisions, and discover actionable knowledge**.
Key Subfields:

- **Data Mining**: **Focused on pattern discovery and predictive modeling through** AI, ML, statistics, and database integration;
- **Data Analytics**: **Broader analysis encompassing visualization**, **metrics**, and interpretive models for business decision-making.

Process Workflow:

1. Define data requirements
2. Collect and preprocess data
3. Clean and transform datasets
4. Perform exploratory data analysis (EDA)
5. Model and apply algorithms
6. Deliver a data product
7. Communicate findings effectively

1.1.3 Applications of Data Analysis

Decision Support Domains:

- **Market Analysis & Management**: Target marketing, CRM, segmentation, cross-selling, and behavioral clustering
- **Risk Management**: Forecasting, quality control, underwriting, and competitive strategy
- **Fraud Detection**: Identifying anomalous or deceptive patterns via outlier and clustering techniques
- **Emerging Areas**: Text mining, web analytics, stream processing, and bioinformatics

1.1.4 Knowledge Discovery in Databases (KDD) Process

A structured sequence for extracting meaningful information:

1. **Domain Understanding**: Define objectives and relevant prior knowledge
2. **Data Selection & Cleaning**: Identify target datasets and perform preprocessing (can consume up to 60% of effort)
3. **Data Transformation & Reduction**: Extract useful features, reduce dimensionality
4. **Mining Function Selection**: Choose tasks—summarization, classification, regression, clustering, or association
5. **Algorithm Selection & Pattern Search**: Execute pattern discovery
6. **Pattern Evaluation & Visualization**: Validate, interpret, and present findings
7. **Deployment**: Integrate insights into operational systems

1.1.5 Data Types in Analysis

- **Traditional**: Relational databases, warehouses, transactional systems.
- **Advanced**: Temporal, sequence, graph, spatial, multimedia, text, web, and heterogeneous data. This diversity necessitates multi-modal analytical frameworks.

1.1.6 Classification

Functional Dimensions:

- **Descriptive vs. Predictive techniques**
- **By Data Type**: Structured, unstructured, streaming, spatial, or web-based
- **By Knowledge Objective**: Association, classification, clustering, trend analysis
- **By Application Domain**: Retail, finance, telecommunications, bioinformatics, web

Course 1.2 ⇒ Data Preprocessing

1.2.1 Overview

Definition:

> **Data preprocessing is the foundational stage of data analysis involving cleaning, transforming, integrating, reducing, and discretizing data to enhance quality and analytical reliability**.

Objective:

> **Transform raw, inconsistent, or noisy datasets into structured, high-quality inputs for machine learning and data mining processes**.

Primary Components:

- Descriptive Data Summarization
- Data Cleaning
- Data Integration & Transformation
- Data Reduction
- Discretization & Concept Hierarchy Generation

1.2.2 Importance of Preprocessing

Real-world data is frequently incomplete, noisy, or inconsistent. **Preprocessing mitigates these issues to ensure that subsequent mining produces accurate, valid, and meaningful results**.

Dirty Data Sources:

- Missing attributes (e.g., incomplete records)
- Noisy measurements (e.g., incorrect salary = “-10”)
- Inconsistent codification (e.g., rating scales “1,2,3” vs. “A,B,C”)
- Duplicate entries or schema discrepancies

Data Quality Dimensions:

- Accuracy: correctness of data values
- Completeness: absence of missing data
- Consistency: coherence across sources
- Timeliness: data relevancy in time
- Believability: perceived credibility
- Value Added, Interpretability, Accessibility

1.2.3 Major Preprocessing Tasks

A. Data Cleaning

Objective: Detect and correct errors to improve reliability.

Core Tasks:

- Fill missing values (mean, median, probabilistic inference)
- Identify and smooth noisy data
- Resolve inconsistencies and redundancy
- Detect and remove outliers

Techniques:

- **Binning**: Smooth data via equal-width or equal-frequency partitions
- **Regression**: Fit data into trend functions for error reduction
- **Clustering**: Group data to isolate anomalies
- **Manual Validation**: Combined computer-assisted inspection for edge cases

B. Data Integration

Definition: **Merge multiple heterogeneous sources into a coherent dataset**.

Challenges:

- Schema Integration: Harmonize field naming (e.g., cust-id ≡ cust-no)
- Entity Identification: Resolve identity conflicts (e.g., “Bill Clinton” = “William Clinton”)
- Redundancy Handling: Detect duplications through correlation or functional dependencies

Outcome: Unified, consistent, and deduplicated dataset ready for modeling.

Statistical Methods:

- Correlation Analysis (Numeric): Pearson’s coefficient rA,Br_{A,B}rA,B for linear relationships.
- Correlation Analysis (Categorical): Chi-square (χ^2) tests for variable dependence.

C. Data Transformation

Purpose: Enhance compatibility and improve analytical accuracy.
Techniques:

- Smoothing: Eliminate noise
- Aggregation: Create higher-level summaries or cubes
- Generalization: Replace low-level data with higher-level abstractions
- Normalization: Rescale data to a standard range
    - Min-Max Normalization: Rescales between predefined bounds.
    - Z-score Normalization: Centers data using mean (μ) and standard deviation (σ).
    - Decimal Scaling: Normalizes via power of 10 adjustment.
- Feature Construction: Derive new attributes that improve model representation.

D. Data Reduction

Goal: Compress datasets while preserving analytical fidelity.
Approach: Obtain a smaller data representation without sacrificing interpretability or statistical integrity.
Key Strategies:

1. Dimensionality Reduction
    - Feature Subset Selection: Remove irrelevant attributes using heuristic selection (forward, backward, or hybrid)
    - Principal Component Analysis (PCA): Identify orthogonal features capturing maximum variance
    - Wavelet Transform: Multi-resolution signal compression (localized and lossy)

2. Numerosity Reduction
    - Regression Models: Fit linear or nonlinear functions to data
    - Log-linear Models: Model multidimensional distributions via lower-order approximations
    - Histograms: Aggregate data into bins using equal-width, equal-depth, V-optimal, or MaxDiff strategies
    - Clustering: Represent datasets through cluster centroids or prototypes
    - Sampling: Use representative subsets (simple, stratified, or cluster-based)

Benefit: Enhances computational efficiency in large-scale analytical workflows.

E. Discretization & Concept Hierarchy Generation

Objective: Convert continuous numerical attributes into categorical intervals, simplifying analysis and improving interpretability.

Attribute Types:

- Nominal: Unordered (e.g., colors, professions).
- Ordinal: Ordered (e.g., ranks, grades).
- Continuous: Real-valued (e.g., age, salary).

Discretization Techniques:

- Unsupervised:
    - Binning and Histogram Analysis (equal-width or equal-depth).
    - Clustering (top-down or bottom-up).
    - χ^2-based Interval Merging (bottom-up).
    - Natural Partitioning (based on local data distributions).

- Supervised:
    - Entropy-based Discretization: Uses information gain to determine optimal split boundaries recursively.

Concept Hierarchy Formation:
Aggregates low-level numerical values into abstract levels (e.g., age → {young, middle-aged, senior}) to facilitate semantic interpretation and model simplification.

1.2.4 Key Insights

- Preprocessing accounts for over 60% of total data mining effort, yet it is indispensable for ensuring analytical integrity.
- Quality data yields quality decisions, preprocessing directly affects mining accuracy, model performance, and interpretability.
- Integration, cleaning, and transformation represent the core backbone of data warehousing pipelines.

Course 1.3 ⇒ Core Concepts

1.3.1 Technique vs Method vs Methodology

- **Technique** ⇒ a systematic procedure, formula, or a routine by which a task is accomplished / a procedure or skill for completing a specific task;
- **Method** ⇒ a habitual, logical, or prescribed practice or systematic process of achieving certain end results with accuracy and efficiency, usually in a preordained sequence of steps / a way something is done;
- **Methodology** = the study of methods used in a field; a collection of methods, practices, procedures, and rules used in some field.

Notice:

Model ⇒ **a simple representation of a system or process, especially one that can be used in calculations or predictions of what might happen**.

1.3.2 Detection vs Prediction vs Forecast

- **Detection** ⇒ Smoke detectors issue warning signals of an impending fire hazard. They don't predict the possibility of a fire accident. Based on early warning, we are presented options: whether to extinguish the fire/smoke source or escape the scene
- **Prediction** ⇒ Prediction is more about understanding consumer behaviour. It is subjective, arbitrary and fatalistic by nature.
- **Forecast** ⇒ Forecasting is all about the numbers and using level and trend and seasonality observations to predict outcomes. It is based on temporal information, scientific and free from intuition and personal bias

1.3.3 Robust & Robustness

- **Robust system** ⇒ strong and unlikely to break or fail
- **Robustness** ⇒ the ability of a computer system to cope with errors during execution and cope with erroneous input
- **Robust programming** ⇒ a programming style that focuses on handling unexpected termination and unexpected actions
- **Robust machine learning algorithm** ⇒ either the testing error has to be consistent with the training error, or the performance is stable after adding some noise to the dataset

Course 2

Course 2.1 ⇒ Fuzzy Sets

2.1.1 Fuzzy Logic vs. Crisp Logic

Definition:

> **Crisp Logic:** ***Traditional (crisp) logic defines truth in binary terms, true or false (1 or 0).***
> **Fuzzy Logic:** ***introduces degrees of truth within the range [0, 1], enabling representation of imprecise, vague, or linguistic information closer to human reasoning***.

Example:

- **Crisp**: "Apply brake at 20m from red light."
- **Fuzzy**: "Slowly apply brake when approaching the red light."

2.1.2 Foundations of Fuzzy Set Theory

Origin:

Formulated by Lotfi A. Zadeh (1965) to handle vagueness and partial membership.

Key Concepts:

- **Membership Function (μA(x))**:
  Defines the degree to which element x belongs to set A, where
  - ***Crisp sets: μA(x) ∈ {0,1} (0 or 1)***
  - ***Fuzzy sets: μA(x) ∈ [0,1] (between 0 and 1)***
- **Support(A)**: All elements where μA(x) > 0
- **Supremum** (**Sup**): Maximum membership value (usually = 1)
- **Infimum** (**Inf**): Minimum membership value (usually = 0)
- **Scalar Cardinality**: Sum of all membership degrees in finite sets, representing "fuzzy count."

2.1.3 α-Cuts and Level Sets

α-Cut Definition: **Transforms fuzzy sets into crisp subsets**

- $\alpha A = \{x \mid \mu A(x) \geq \alpha\}$ — standard α-cut
- $\alpha^{+}A = \{x \mid \mu A(x) > \alpha\}$ — strong α-cut

Produces a hierarchy of nested sets: $\boldsymbol{\alpha_1 A \subseteq \alpha_2 A \subseteq \alpha_3 A \subseteq \ldots}$

2.1.4 Convexity and Fuzzy Subsets

Convex Fuzzy Set:

For any x, y in X and $0 \leq \lambda \leq 1$, $\mu A(\lambda x + (1 - \lambda)y) \geq \min(\mu A(x), \mu A(y))$

Interpretation:

All α-cuts of a convex fuzzy set are convex regions.

Convexity ensures smoothness and stability in fuzzy reasoning and arithmetic.

Subset Relationship:

$\boldsymbol{A \subseteq B \Leftrightarrow \mu A(x) \leq \mu B(x) \ \forall x \in X}$

2.1.5 Advanced Fuzzy Set Types

- **Interval-Valued Fuzzy Set**: Membership degree represented as an interval [l,u] within [0,1], to handle uncertainty in μA(x)
- **Type-2 Fuzzy Set**: Each membership value itself is a fuzzy set within [0,1], enabling modeling of uncertainty about uncertainty

2.1.6 Constructing Fuzzy Sets

Sources:

- Direct expert definitions (linguistic or numeric)
- Indirect pairwise comparisons
- Learning from data (data-driven fuzzification)

Linguistic Variables:

Express qualitative concepts numerically.

2.1.7 Fuzzy Set Operations

Standard Zadeh Operators (these generalize Boolean algebra to fuzzy domains)

- **Complement**: $\mu A^c(x) = 1 - \mu A(x)$
- **Intersection (AND)**: $\mu A \cap B(x) = \min(\mu A(x), \mu B(x))$
- **Union (OR)**: $\mu A \cup B(x) = \max(\mu A(x), \mu B(x))$

Preserved Properties:

- **Involution**: $(A^C)^C = A$
- **Commutativity**: $A \cap B = B \cap A$
- **Associativity**: $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributivity**: $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
- **Idempotence**: $A \cap A = A$, $A \cup A = A$
- **Absorption**: $A \cap (A \cup B) = A$, $A \cup (A \cap B) = A$, $A \cup X = X$, $A \cap \Phi = \Phi$
- **Identity**: $A \cap X = A$, $A \cup \Phi = A$
- **De Morgan's Laws**: $(A \cup B)^C = (A^C \cap B^C)$ / $(A \cap B)^C = (A^C \cup B^C)$

Unpreserved Properties:

- **Law of excluded middle**: $A \cup A^C = X$
- **Law of contradiction**: $A \cap A^C = \Phi$

2.1.8 Generalized Operator Classes

Fuzzy Complement Functions
Axioms:

- Boundary: $c(1)=0$, $c(0)=1$
- Monotonicity: $a \le b \Rightarrow c(a) \ge c(b)$
- Continuity & Involution: $c(c(a))=a$

Notable Classes:

- Zadeh: $c(a)=1-a$
- Sugeno: cλ(a)=1−a1+λac_λ(a)=\frac{1−a}{1+λa}cλ(a)=1+λa1−a, λ > –1
- Yager: cw(a)=(1–aw)1/wc_w(a)=(1–a^w)^{1/w}cw(a)=(1–aw)1/w, w > 0
  These allow “optimistic” or “pessimistic” complements depending on parameterization

2.1.9. Fuzzy Intersection (T-Norms)

Core Axioms:

- **Boundary**: $i(a,1) = a$
- **Monotonicity**: $b<=d \Rightarrow i(a,b) <= i(a,d)$
- **Commutativity**: $i(a,b) = i(b,a)$
- **Associativity**: $i(a,i(b,d)) = i(i(a,b),d)$
- **Subidempotency**: $i(a,a) < a$
- **Strict monotonicity**: $a_1<a_2$ and $b_1<b_2 \Rightarrow i(a_1,b_1) < i(a_2,b_2)$

Examples:

- Zadeh (min): $i(a,b)=\min(a,b)$
- Algebraic Product: $i(a,b)=ab$
- Bounded Difference: $i(a,b)=\max(0, a+b-1)$
- Drastic: $i(a,b)=a$ if $b=1$; otherwise 0
- Yager Family: covers all between imin(a,b) and min(a,b).

2.1.10 Fuzzy Union (T-Conorms)

Core Axioms: **Boundary**, **commutativity**, **associativity**, **monotonicity**.

Examples:

- Zadeh (max): $u(a,b)=\max(a,b)$
- Algebraic Sum: $u(a,b)=a+b-ab$
- Bounded Sum: $u(a,b)=\min(1, a+b)$
- Drastic: $u(a,b)=1$ if either input $\neq 0$.
- Yager Family: parameterized transitions between max and umax.

2.1.11. Duality and Operator Triples

De Morgan's Laws: $(A \cup B)^C = (A^C \cap B^C)$ / $(A \cap B)^C = (A^C \cup B^C)$

A dual triple $\langle i, u, c \rangle$ satisfies De Morgan's laws:

$$c(u(a,b)) = i(c(a), c(b))$$

$$c(i(a,b)) = u(c(a), c(b))$$

This ensures logical consistency between union, intersection, and complement operations.

2.1.12 Aggregation Operators

Definition:

Generalized operators combining multiple fuzzy values into one: $[0,1]^n \Rightarrow [0,1]$

Axioms:

- Boundary ($0 \to 0$, $1 \to 1$), monotonicity, idempotency, symmetry.

Families:

- **Mean Operators**: Arithmetic, geometric, harmonic, min, max
- **Generalized Mean**: Parameter α defines mean type (e.g., α=1 arithmetic, α=0 geometric)
- **Weighted Means**: Apply attribute weights
- **OWA** (**Ordered Weighted Averaging**): Weights depend on sorted input order, models linguistic terms like most, some, at least k of n
- **Hybrid Operators**: Combine t-norms and t-conorms via additive or multiplicative weighting

2.1.13 Fuzzy Numbers and Arithmetic

Fuzzy Number:

A convex, normal fuzzy set on $\mathbb{R}$ representing an approximate numeric quantity (e.g., "around 2").

Properties:

- Normal (max membership = 1)
- Convex (each α-cut is an interval)

Arithmetic via α-Cuts (for fuzzy numbers A, B):

- **$\alpha(A \oplus B) = \alpha A + \alpha B$**
- **$\alpha(A \ominus B) = \alpha A - \alpha B$**
- **$\alpha(A \otimes B) = \alpha A \times \alpha B$**
- **$\alpha(A \div B) = \alpha A \div \alpha B$ (if $0 \notin \alpha B$)**

2.1.14. Extension Principle

Definition:

General method to extend a crisp function y = f(x) to fuzzy sets:
$\mu B(y) = \sup [\mu A(x) \mid f(x)=y]$

Purpose:

Allows arbitrary functions and operations (including multi-variable ones) to be applied to fuzzy inputs.

Applications:

- Fuzzy arithmetic (addition, subtraction, multiplication)
- Fuzzy function evaluation (e.g., sin(A))
- General fuzzy transformations (mappings, models)

2.1.15 Core Insights

- Fuzzy sets generalize classical logic, enabling reasoning under uncertainty.
- Membership functions encode subjective or linguistic concepts numerically.
- Fuzzy operators (complements, t-norms, t-conorms) define algebraic manipulation of vagueness.
- The Extension Principle unifies fuzzy arithmetic and function transformation.
- Fuzzy numbers and α-cuts form the computational foundation for fuzzy control and inference systems.

Course 3

Course 3.1 ⇒ Cluster Analysis

3.1.1 Classification vs. Clustering

Classification (Supervised Learning):

- **Learns a mapping function from labeled training data**
- Input: pairs of (features, labels)
- Output: an inferred function that predicts class labels for unseen instances
- Objective: Generalize effectively from known to unknown data
- Quality Measures: Accuracy, precision, recall

Clustering (Unsupervised Learning):

- **No labels provided**: the algorithm infers structure or grouping
- The accuracy of clusters cannot be objectively measured against known classes
- Core purpose: **discover hidden patterns**, natural groupings, or similarity structures within unlabeled data

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3.1.2. Algorithmic Concepts

- **Soundness**: Every returned result is a valid solution.
- **Completeness**: Every possible valid solution can be found by the algorithm.
- **Accuracy**: **% of all correct predictions**
- **Precision**: **% of predicted positives that are actually positive**
- **Recall**: **% of actual positives correctly identified**

3.1.3. Fundamental Definitions

Cluster Analysis (Clustering):

> **The process of grouping data points such that members within a cluster are more similar to each other than to those in other clusters**.
> **It is a core task in exploratory data mining**.

Key Characteristics:

- No universal definition of a “cluster.”
- The process is iterative, involving parameter tuning, preprocessing adjustment, and interpretation.
- The “right” clustering depends on data structure, objective, and similarity measure.

3.1.4. Cluster Types and Membership Models

- **Hard (Crisp) Clustering**: Strict partitioning clustering, means each object belongs to exactly one cluster
  Strict partitioning clustering with outliers: objects can also belong to no cluster
- **Soft (Fuzzy) Clustering**: Objects belong to clusters with degrees of membership
- **Partitioning Clustering**: Strict division into non-overlapping groups
- **Hierarchical Clustering**: Nested clusters forming a tree-like structure
- **Overlapping Clustering**: Objects can belong to multiple clusters
- **Subspace Clustering**: Clusters defined within specific attribute subsets

Course 3.2: Core Cluster Models

Model Type	Concept	Representative Algorithms
Connectivity (Hierarchical)	Clusters formed by linking nearby points based on distance	Single-Link, Complete-Link, UPGMA
Centroid Model	Each cluster represented by a center (mean or medoid)	k-Means, k-Medoids, k-Means++
Distribution Model	Clusters as probabilistic distributions	EM Algorithm, Gaussian Mixture Models
Density Model	Clusters as high-density regions separated by sparse areas	DBSCAN, OPTICS, Mean-Shift
Subspace Model	Clusters defined by both members and attribute subsets	Biclustering, Co-Clustering
Graph Model	Clusters as cliques or quasi-cliques in graphs	HCS Clustering
Neural Model	Clustering via self-organizing maps or neural encoding	SOM, PCA/ICA-based clustering

3.2.1 Hierarchical (Connectivity-Based) Clustering

Concept:

Objects are grouped based on pairwise proximity — closer objects merge first

Variants:

- Agglomerative: Start with individual points and merge upwards
- Divisive: Start with one cluster and recursively divide

Linkage Criteria:

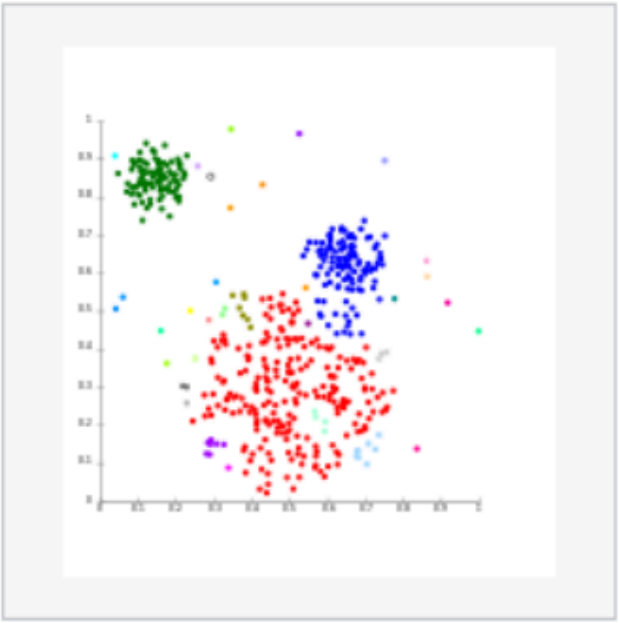
- Single-Linkage: Minimum distance between cluster members
- Complete-Linkage: Maximum distance between cluster members
- Average-Linkage (UPGMA): Mean pairwise distance

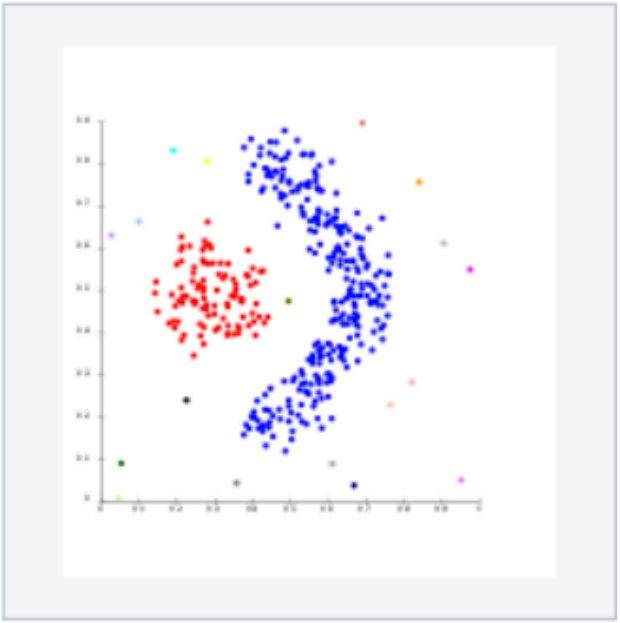
Output:

Dendrogram — a tree showing hierarchical relationships

Limitations:

- Sensitive to outliers ("chaining effect")
- Computationally intensive ($O(n^3)$ for agglomerative)
- Not ideal for large or noisy datasets





3.2.2 Centroid-Based Clustering

Concept:

Represent clusters by central points minimizing within-cluster variance

K-Means Algorithm:

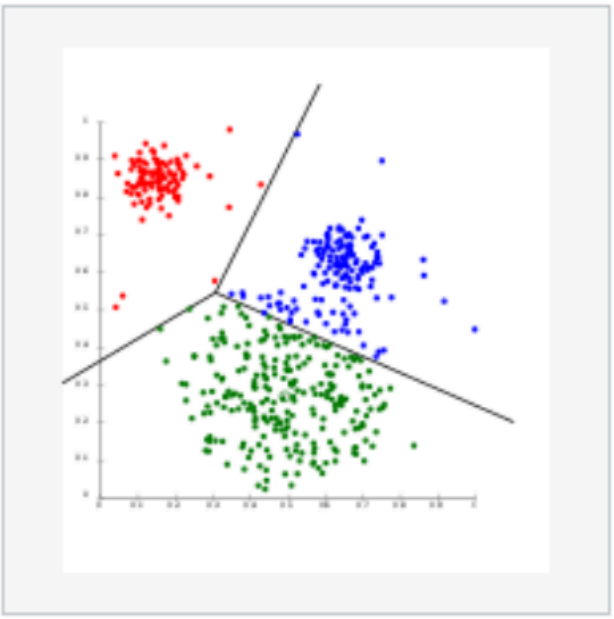
1. Initialize k cluster centers
2. Assign points to the nearest center
3. Update centers based on assignments
4. Repeat until convergence

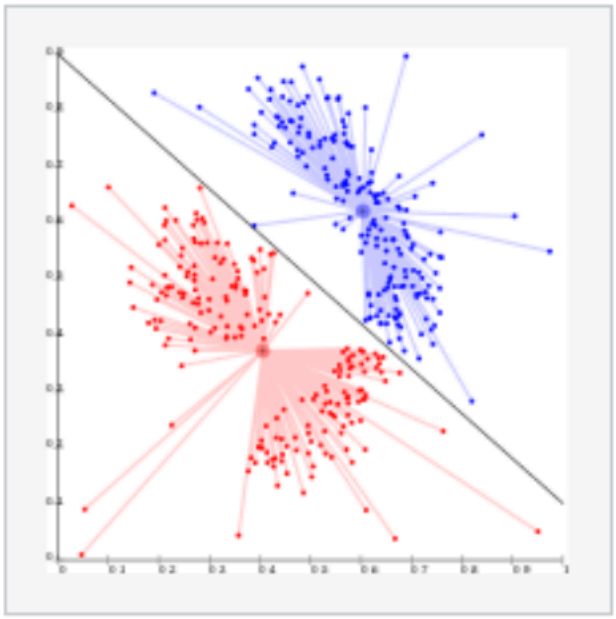
Properties:

- Produces Voronoi partitions of space
- Efficient and conceptually linked to nearest-neighbor classification
- Requires predefined k
- Sensitive to initialization and assumes convex, spherical clusters

Variants:

- k-Medoids: Uses actual data points as centers
- k-Medians: Uses median distance measures
- k-Means++: Improves initial center selection
- Fuzzy C-Means: Allows soft membership in multiple clusters





3.2.3 Distribution-Based Clustering

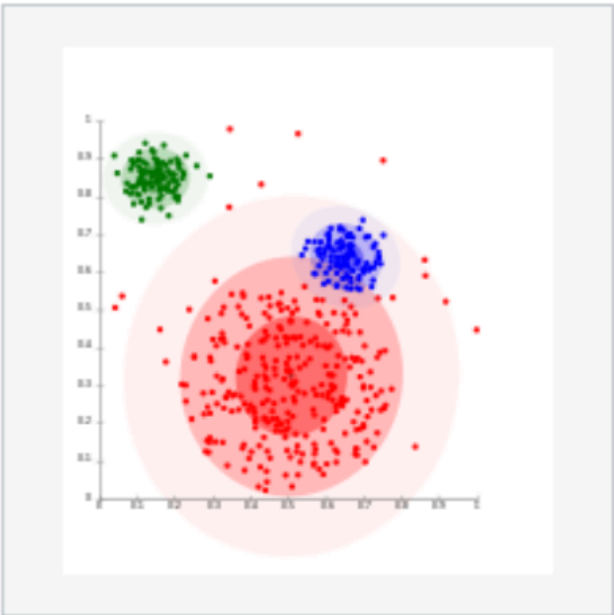
Concept:
Assumes data points are generated by a mixture of probability distributions

Common Model:
Gaussian Mixture Models (GMMs)

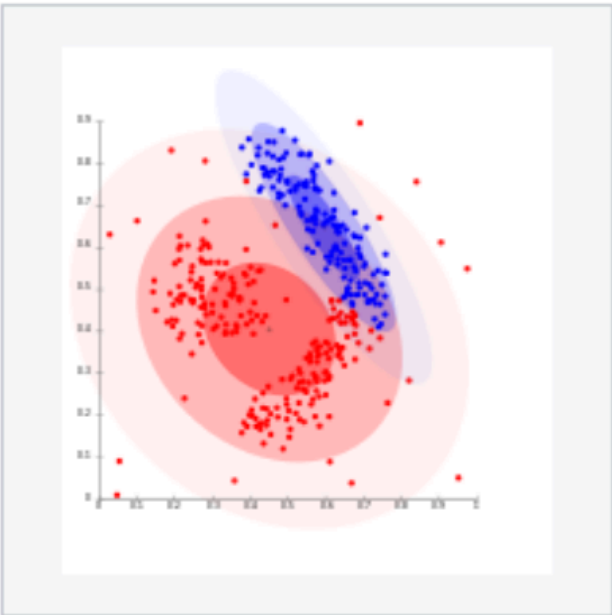
Method:
- Model data as k Gaussian distributions.
- Estimate parameters via the Expectation-Maximization (EM) algorithm.
- Soft assignments (probabilistic memberships) or hard assignments via maximum likelihood.

Advantages:
Handles overlapping clusters and captures statistical dependencies.

Drawbacks:
Sensitive to initialization, may overfit without regularization.



On Gaussian-distributed data, EM works well, since it uses Gaussians for modelling clusters



Density-based clusters cannot be modeled using Gaussian distributions

3.2.4 Density-Based Clustering

Concept:

Clusters are dense regions of points separated by areas of lower density

Key Algorithm — DBSCAN:

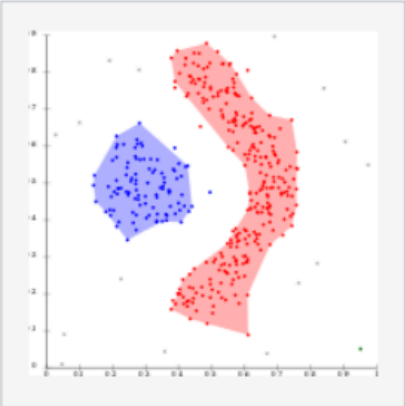
- Defines clusters by ε (radius) and MinPts (minimum points)
- Connects points that are "density-reachable"
- Detects arbitrary-shaped clusters and noise points
- Deterministic and efficient (O(n log n) with spatial indexing)

Variants:

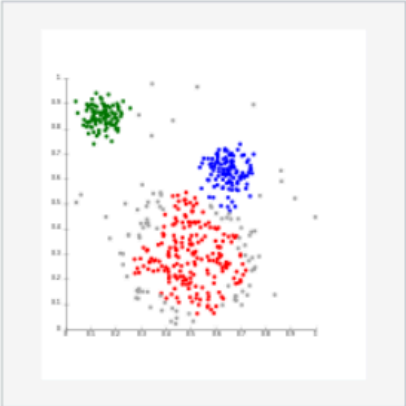
- OPTICS: Removes dependency on fixed ε, produces hierarchical clustering
- DeLi-Clu: Combines OPTICS with linkage-based clustering for improved speed
- Mean-Shift: Moves points iteratively to local density maxima (kernel-based)

Limitations:

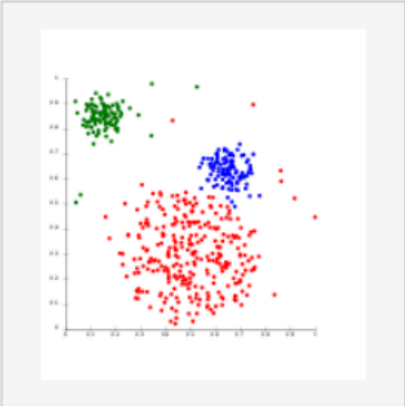
- Fails on overlapping Gaussian clusters (gradual density transitions)
- Parameter sensitivity (choice of ε and MinPts)



Density-based clustering with DBSCAN.



DBSCAN assumes clusters of similar density, and may have problems separating nearby clusters



OPTICS is a DBSCAN variant that handles different densities much better

3.3 Fuzzy clustering ⇒ LINK

Data representation:
objects are vectors of measured values
Feature extraction:
to reduce data dimensionality and eliminate redundant characteristics
Clusters shape:
different geometric prototypes; norms or scalar products
Clusters size:
use of adaptive distance or adaptive algorithms
Clusters validity:
optimal number of classes through validity functionals, clusters merging/splitting or by using a hierarchical approach
Final fuzzy partition:
needs to be defuzzified; it should not be discarded
Method:
fuzzy objective function minimization; two step iterative procedure that continually decreases the value of the objective function

Function:

- Given: c, n, m, and x^j, j = 1_.._n; l = 0;

$$A_i^{(l+1)}(x^j) = \frac{1}{\sum_{k=1}^{c} \left(\frac{D(x^j, L_i)}{D(x^j, L_k)} \right)^{\frac{1}{m-1}}}$$

3.3.2 Fuzzy Incomplete Data Clustering

- **Objective**: **Perform clustering with missing or incomplete data**.
- Methods:
  - Whole Data Strategy: Removes incomplete items.
  - Partial Distance Strategy: Uses only available features.
  - Optimal Completion: Iteratively estimates missing values via membership-weighted averages.
  - Nearest Neighbour: Replaces missing values using nearest prototype estimates.

3.3.3 Fuzzy Shell Clustering

Purpose: **Identify shell-like (non-solid) structures such as circles, ellipses, or surfaces**.

3.3.4 Fuzzy Clustering Models

Hierarchical and Cross Models:

- Divisive Hierarchical Clustering: Builds binary fuzzy hierarchies.
- Simultaneous (Cross) FCM: Clusters both observations and variables iteratively (biclustering).
- Associative/Hierarchic Cross-Clustering: Enhances interpretability using fuzzy correlation matrices.

Fuzzy Biclustering: Produces fuzzy biclusters with adaptive metrics for richer representation.

3.3.5 Global Quality Measures

Objective: **Used to evaluate clustering validity and separation**.

Solid Clustering Metrics

- Partition Coefficient (PC), Entropy (PE), Normalized Variants (NPC, NPE): Measure fuzziness and clarity of clusters.
- Xie–Beni Index: Ratio of compactness to separation (to minimize).
- Backer–Jain Index: Measures cluster discrimination (to maximize).
- Separation and Dunn Indices: Assess compactness and inter-cluster distance.

Shell Clustering Metrics

- Fuzzy Hypervolume (FHV): Measures spread of clusters.
- Partition Density (PD), Average Partition Density (APD): Assess density within shells.
- Shell Thickness: Quantifies geometric tightness of shell clusters.

3.3.6 Dimensionality Reduction

Principal Component Analysis (PCA): Reduces dimensionality using eigenvalues and eigenvectors of covariance matrices.
Variable Selection: Based on eigenvalues, loadings, or communalities.
Multidimensional Scaling (MDS): Projects data into lower dimensions preserving distances (metric/non-metric variants).
Sammon Mapping: Preserves local structure during nonlinear projection.

3.3.7 Supervised Classification

Perceptron-based fuzzy classifiers: Linear separations adapted for fuzzy partitions.
Keller–Hunt and Gallant Pocket Algorithms: Enhance robustness and convergence.
Fuzzy Relaxation, Minimal Square Error, Widrow–Hoff, Ho–Kashyap: Advanced fuzzy linear classifiers.
Fuzzy k-Nearest Neighbors, Nearest Prototype, Restricted FCM: Nonlinear and adaptive classification methods.

3.3.8 Regression

Fuzzy Regression: Models uncertain relationships with fuzzy distances and membership-based weighting.
Least Squares and Polynomial Fits: Standard and fuzzy-enhanced regression models.
Nonlinear Regression (Levenberg–Marquardt): For complex nonlinear relationships.

3.3.9 Regression Quality Measures

(Not fully detailed in the provided section but generally include error-based and membership-weighted validation indices).

Course 4 ⇒ LINK

4.1.1 Regression

Concept:

Model relationships between dependent and independent variables under uncertainty and data imperfections.

Traditional (Crisp) Regression:

- Employs Least Squares (LS) minimization
- Assumes homoscedasticity (constant variance) and clean data
- Sensitive to outliers and heteroscedasticity

Motivation for Fuzzy Regression:

- Incorporates data uncertainty through fuzzy membership degrees
- Enables robust modeling under imprecision or incomplete data
- Integrates fuzziness as adaptive data weights in regression modeling
- Fuzzy Regression Algorithm:
    - Defines a fuzzy set A and a prototype L minimizing:
    - Iteratively optimizes both membership degrees and prototype location.
    - α defines partition polarization; typical α ≈ 0.10 ensures robust performance.

Fuzzy Linear Regression (FLR):

- Employs distance to line rather than vertical distance
- Prototype determined via fuzzy mean and principal component
- Produces robust, adaptive, and scale-independent fits superior to traditional methods

4.1.3 Principal Components Analysis (PCA)

Purpose:

Reduce data dimensionality by constructing new orthogonal variables (principal components) maximizing explained variance.

Process:

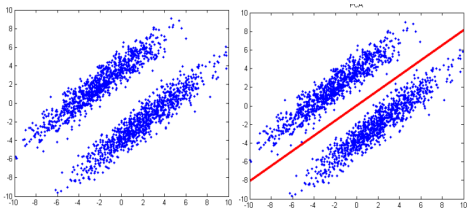
- Compute covariance or correlation matrix
- Extract eigenvalues/eigenvectors (principal components and variances)
- Project data onto reduced space $X' = X \cdot E$ $X' = X \cdot E$ $X' = X \cdot E$

Fuzzy PCA:

- Introduces fuzzy weights (membership degrees) to handle isolated data and outliers
- Employs fuzzy covariance matrix:
- Enhances robustness by reducing the influence of atypical data points.

Component Selection Criteria:

- Eigenvalue magnitude (>1 rule).
- Consecutive eigenvalue difference.
- Cumulative variance ratio threshold (e.g., >90%).



4.1.4 Multi-Dimensional Scaling (MDS)

Objective:

Represent proximity relationships between data points in a lower-dimensional metric space.

Metric MDS (Shepard, 1962):

Minimizes Euclidean distance deviations between original and projected distances.

Non-Metric MDS (Kruskal, 1964):

Uses monotonic transformations preserving rank order of distances.

Sammon Mapping (1969):

Focuses on preserving local distances, yielding more interpretable spatial layouts.